

DocClaw: A Unified Agentic System for Intelligent Document Processing

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Abstract

Intelligent document processing (IDP) encompasses a broad range of tasks, including optical character recognition (OCR), document question answering (DocQA), and key information extraction (KIE). Despite their distinct objectives, these tasks share a common need to perceive document content, acquire task-relevant information, and progressively refine intermediate results. However, they are typically formulated as separate prediction problems and addressed by task-specific models or processing pipelines. We introduce DocClaw, a unified agentic system that formulates diverse intelligent document processing tasks as a shared process of interaction between an agent and a document. Given a document and a task-specific query, DocClaw follows an appropriate document skill to iteratively identify the information required, invoke relevant tools, and integrate the resulting observations into the desired output. Throughout this process, a structured document state organizes reusable document knowledge and task-specific interaction context, allowing the agent to accumulate, revisit, and progressively refine information as the interaction proceeds. Under this formulation, task-specific requirements are captured by the agent’s interpretation of the query objective and the corresponding document skill, while the underlying interaction loop, tool space, and document state are shared across tasks. Extensive experiments across multiple intelligent document processing benchmarks demonstrate that DocClaw effectively handles diverse tasks within a single agentic framework and achieves competitive performance compared with both general-purpose VLMs and task-specific methods.

1 Introduction

Intelligent document processing (IDP) aims to extract and reason over information jointly conveyed through textual content, spatial layout, and visual elements (Appalaraju et al. 2021; Li et al. 2021b; Tang et al. 2023). It encompasses a broad spectrum of tasks, ranging from fundamental document perception tasks such as optical character recognition (OCR) to higher-level document understanding tasks such as document question answering (DocQA) and key information extraction (KIE) (Li et al. 2021a; Xu et al. 2020; Ji et al. 2026). Existing IDP methods are predominantly developed around individual task objectives, resulting in separate processing chains for different document processing tasks. As illustrated in Figure 1, these methods typically follow one of two processing paradigms. Some methods adopt single-pass

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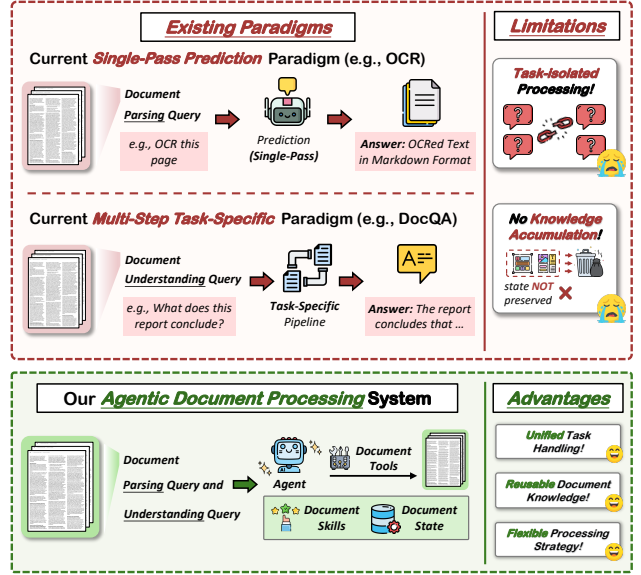


Figure 1: Unlike existing single-pass and multi-step paradigms, the proposed DocClaw formulates intelligent document processing as interactions between an agent and a document, leveraging iterative tool invocation and knowledge accumulation across queries.

prediction, directly producing task-specific outputs without intermediate processing steps (Ye et al. 2023; Xu et al. 2026; Dong et al. 2026). Others rely on multi-step task-specific pipelines, which decompose document processing into predefined sequences of specialized operations, progressively transforming intermediate results into the required outputs. By separating layout analysis from region-level recognition, these pipelines have achieved promising performance in document parsing (Feng et al. 2025; Duan et al. 2026; Zhang et al. 2025, 2026b; Wang et al. 2026).

Despite these advances, most existing work treats intelligent document processing as a collection of isolated tasks, leading to two major limitations. First, there remains an absence of a general framework for handling heterogeneous document processing tasks within a unified workflow. Single-pass models rely on rigid, task-specific mappings, while multi-step pipelines remain constrained by predefined task-

specific procedures. As a result, neither paradigm can flexibly adapt its processing pattern across diverse intelligent document processing tasks. Second, processing each task independently hinders document knowledge from being accumulated, reused, and refined across queries. Document processing often involves interdependent workloads, where intermediate results can naturally serve as reusable knowledge for subsequent operations. However, isolated task processing provides no unified mechanism for sharing such knowledge.

To address the above limitations, we introduce DocClaw¹, a unified agentic document processing system that reformulates document processing as iterative interactions with documents. Through an iterative interaction loop, DocClaw actively explores document content, selects and invokes appropriate document processing tools, updates the document state with newly acquired knowledge, and progressively refines intermediate results into task-specific outputs. To support this iterative process, DocClaw maintains a structured document state that serves as the agent’s memory. The document state comprises document memory, which preserves document knowledge for reuse across queries with different objectives, and task memory, which tracks task-specific context throughout the current interaction. To guide the agent’s processing strategy across different query objectives, we introduce document skills, a set of task-oriented instructions that define document interaction strategies and tool-use patterns. For each query, DocClaw selects the skill corresponding to its objective to guide subsequent planning and tool invocation.

We evaluate DocClaw across three representative intelligent document processing tasks: optical character recognition (OCR), document question answering (DocQA), and key information extraction (KIE). Experiments show that DocClaw achieves competitive performance against both general-purpose vision-language models and task-specific methods, demonstrating its effectiveness in supporting heterogeneous document processing tasks within a unified agentic framework. Further ablation studies demonstrate that the document state enables effective accumulation and reuse of document knowledge, whereas document skills provide task-oriented guidance that steers the agent toward task-specific processing strategies, allowing intermediate results to be progressively improved through dedicated refinement operations tailored to different query objectives. Our main contributions are summarized as follows:

- We introduce DocClaw, a unified agentic framework that formulates intelligent document processing as iterative interactions between an agent and a document, allowing different query objectives to be addressed within a shared interaction loop and document processing tool space.
- We propose a structured document state that organizes persistent document knowledge and task-specific context, supporting progressive knowledge accumulation, reuse, and refinement across interactions and queries.
- We introduce document skills, a set of task-oriented instructions that guide the agent toward appropriate processing strategies and tool-use patterns for different query

objectives, together with dedicated refinement operations that progressively improve intermediate results.

2 Related Work

Intelligent Document Processing Intelligent document processing aims to derive structured or semantic outputs from the textual and visual information embedded in a document. Existing intelligent document processing methods can be broadly categorized into single-pass prediction and multi-step task-specific pipelines. Single-pass prediction utilizes specialized models for individual tasks, such as optical character recognition (OCR) and document layout analysis (DLA). These models directly predict task-specific outputs from document images, with their architectures and training objectives designed for corresponding task formulations (Dong et al. 2026; Team et al. 2025). While effective for their target tasks, they are generally difficult to adapt to heterogeneous document processing objectives. Multi-step task-specific pipelines decompose document processing into a sequence of specialized operations. For example, PaddleOCR-VL (Zhang et al. 2026b) first performs layout analysis to identify and order document regions, then recognizes individual regions and assembles their outputs into a complete document transcription. Similar multi-step pipelines have also been explored for document understanding, where systems retrieve document content, inspect candidate pages, and generate task-specific outputs (Sun et al. 2025a; Han et al. 2025; Zhang et al. 2026a). However, their operations are typically organized into predefined chains designed for specific tasks. Reusable capabilities such as layout analysis, text recognition, and information retrieval remain embedded in separate task-specific pipelines. In contrast, DocClaw organizes these capabilities within a unified agentic framework that dynamically plans and invokes document processing tools according to the query objective and document state.

Agentic and Tool-Augmented Reasoning. Agentic and tool-augmented reasoning has emerged as an effective paradigm for extending language models beyond text generation. Retrieval-augmented methods ground model outputs in external evidence, while tool-augmented approaches enable models to invoke external tools and incorporate the resulting observations into subsequent reasoning (Lewis et al. 2020; Yao et al. 2023; Schick et al. 2023). Multi-agent frameworks further extend this interaction paradigm by coordinating agents with different roles and capabilities to collaboratively solve complex tasks (Li et al. 2023; Wu et al. 2023; Chan et al. 2024). Despite these advances, most existing agentic systems are designed for general-purpose reasoning and treat external tools as independent resources invoked to solve a particular task. When applied to document processing, they lack a unified mechanism for coordinating heterogeneous document tools and maintaining reusable document knowledge across interactions. Consequently, document knowledge acquired during parsing, retrieval, and reasoning is difficult to accumulate, refine, and reuse over time. To address this limitation, DocClaw introduces a unified agentic framework that coordinates document tools through an agentic interaction loop and maintains a structured document state to accumulate,

¹Code is available at <https://github.com/sxiangag/DocClaw>.

refine, and reuse document knowledge across interactions.

3 Methodology

This section details our proposed framework, DocClaw. We first formulate intelligent document processing as iterative interactions between an agent and a document. We then present the document state, a structured memory that organizes document knowledge and task-specific interaction context throughout the agentic interaction process. Finally, we introduce document skills, a set of task-oriented instructions that guide the agent’s interaction strategies and tool-use patterns under different query objectives.

3.1 Agentic Document Processing

Formally, given an input document D and a query q , the goal of intelligent document processing is to produce a task-specific output y in response to q . Depending on the query objective, y may take different forms, such as recognized document content for OCR, extracted field-value pairs for KIE, or answers to questions for DocQA. DocClaw provides a unified formulation for these heterogeneous objectives by treating document processing as iterative interactions between an agent and the document, as illustrated in Figure 2.

For each user query q , DocClaw first selects a document skill $k \in \mathcal{K}$ from a predefined skill set according to the query objective. The selected skill specifies the task-oriented interaction strategy and tool-use patterns that guide subsequent processing. The document state is then initialized as $s_0 = (m_0, e_0)$, consisting of document memory m_0 and task memory e_0 . For a previously unseen document, the document memory is initialized from its page images $\{p_i\}_{i=1}^N$, whereas for a document that has already been processed, the retained document memory serves as the starting state for the new query. In contrast, the task memory is initialized independently for each query and starts empty, i.e., $e_0 = \emptyset$.

After initialization, an LLM-based agent interacts with the document state using document processing tools through a ReAct-style (Yao et al. 2023) agentic loop. At each interaction step t , the agent first reads a condensed representation $\phi(s_t)$ of the current document state, which summarizes what document information has already been acquired and what task-specific interaction progress has been recorded. Conditioned on $\phi(s_t)$, the user query q , and the selected document skill k , the planner determines the next document processing action:

$$\begin{aligned} a_t &= (\tau_t, c_t) = \pi_\theta(q, k, \phi(s_t)), \\ \tau_t &\in \mathcal{T}, \quad c_t \in \mathcal{C}(\tau_t), \end{aligned} \quad (1)$$

where τ_t denotes the selected document processing tool, c_t specifies its invocation arguments, and $\mathcal{C}(\tau_t)$ denotes the valid argument space of tool τ_t . This allows the agent to dynamically determine the document operation to perform and its corresponding invocation arguments based on the current document state.

The planned action is then executed on the document and produces an execution result r_t . Rather than being used as a transient observation, the execution result is incorporated into the document state to support subsequent interactions and preserve acquired document knowledge for future

queries. We formulate the tool execution and state update as:

$$\begin{aligned} r_t &= \mathcal{F}_{\tau_t}(D, c_t; s_t), \\ s_{t+1} &= \mathcal{U}(s_t, a_t, r_t) \\ &= (\mathcal{U}_m(m_t, a_t, r_t), \mathcal{U}_e(e_t, a_t, r_t)), \end{aligned} \quad (2)$$

where $\mathcal{F}_{\tau_t}$ denotes the execution of the selected tool τ_t with invocation configuration c_t , and $\mathcal{U}$ denotes the corresponding state update. The update functions $\mathcal{U}_m$ and $\mathcal{U}_e$ integrate reusable document knowledge and task-specific interaction context into document memory and task memory, respectively. Specifically, document knowledge acquired through tool execution, such as recognized content and layout structures, is accumulated in document memory, while task-specific context is recorded in task memory. Through this process, successive tool calls progressively enrich the document state, allowing information accumulated from earlier steps to support subsequent interactions and future queries.

After the state update, the planner reads $\phi(s_{t+1})$ of the updated document state to determine the next document-processing action with reference to the user query q and the selected skill k . The interaction continues until the agent determines that sufficient information has been acquired to satisfy the query and selects an output action a_{out} . For the j -th query $q^{(j)}$ issued on the document D , the final output and the transition to the next query are formulated as:

$$\begin{aligned} a_{T_j}^{(j)} &= a_{\text{out}}, \\ y^{(j)} &= \mathcal{F}_{\text{out}}(q^{(j)}, k^{(j)}, s_{T_j}^{(j)}), \\ m_0^{(j+1)} &= m_{T_j}^{(j)}, \quad e_0^{(j+1)} = \emptyset, \end{aligned} \quad (3)$$

where T_j denotes the final interaction step for query $q^{(j)}$. After the output is generated, the task memory for the completed query is discarded, while the document memory is retained. This allows subsequent queries to reuse previously acquired document knowledge, reducing redundant tool execution while avoiding irrelevant task-specific context.

Through this iterative process, heterogeneous intelligent document processing queries share a unified agentic interaction loop, while task-oriented skills guide tool usage and persistent state updates enable document knowledge to be progressively accumulated, refined, and reused across queries.

3.2 Document State

During iterative agentic interactions, tool execution produces both reusable document knowledge and task-specific context. To organize these two types of information, DocClaw maintains a structured document state:

$$s_t = (m_t, e_t), \quad (4)$$

where m_t denotes persistent document memory and e_t denotes query-level task memory. This separation allows tool-acquired document knowledge to be continuously reused and refined while preventing task-specific context from carrying over to subsequent interactions.

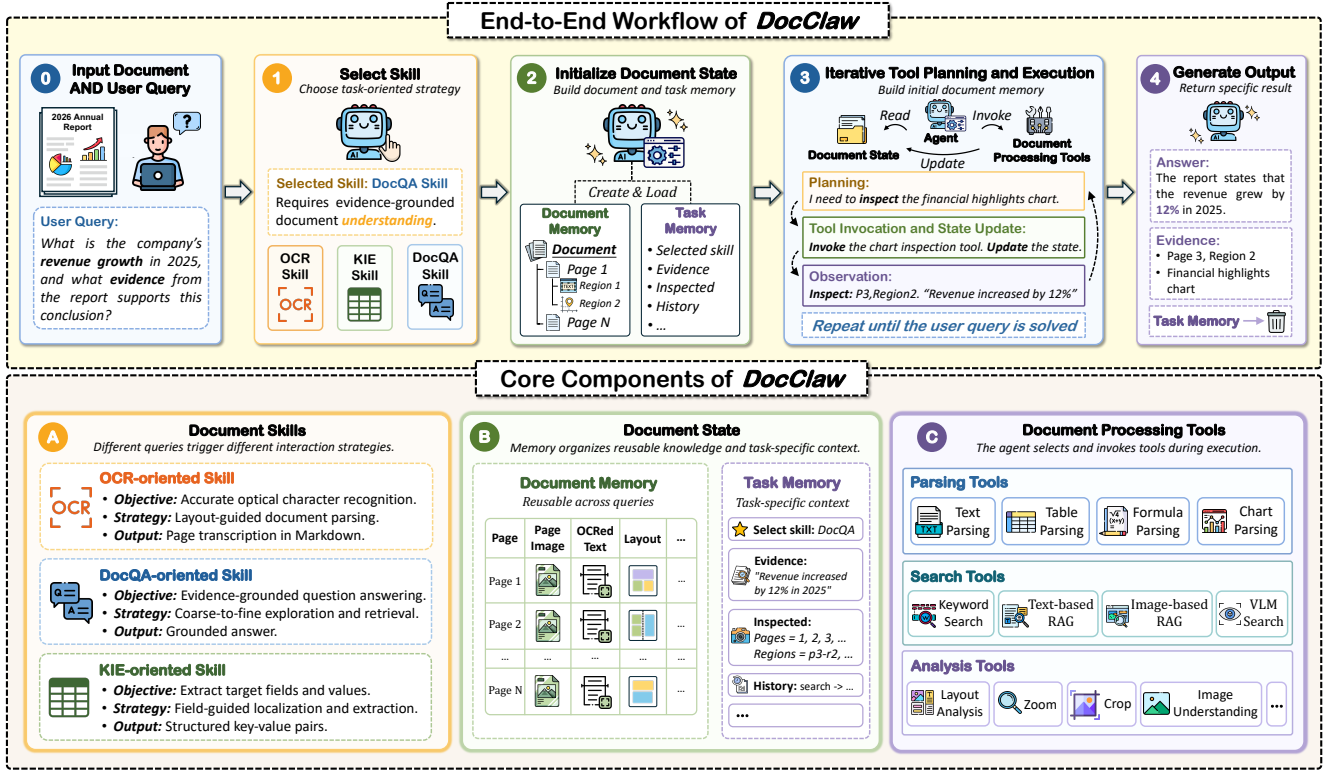


Figure 2: Overview of the proposed DocClaw. Given an input document and a user query, DocClaw first selects a relevant task-oriented document skill and initializes the document state for the current query. It then iteratively plans and executes document processing tools, updating the document state until the final task-specific output is generated.

Document Memory. The document memory m_t organizes document knowledge acquired throughout agentic interactions. To preserve the intrinsic structure of documents, DocClaw organizes this knowledge hierarchically at the document, page, and region levels. The document level maintains global information about the document and its constituent pages. Each page stores its raw page image together with page-level information acquired during processing, such as OCRred text, layout structure, and reading order. When regions are identified, they are associated with their corresponding pages and maintain region-level information, including spatial coordinates, region type, recognized text, and other tool-produced attributes.

This hierarchical organization provides a shared structure through which different document processing tools can progressively add and refine document knowledge. For example, layout analysis identifies page regions and their structural properties, while recognition tools attach recognized content to the corresponding pages and regions. Document knowledge acquired by different tools is therefore consolidated around shared document entities and remains reusable across queries. During planning, the agent accesses the document state through $\phi(s_t)$, which retains only the information necessary for planning and tool selection. Therefore, contents such as page images and recognized text remain accessible to document processing tools while being omitted from the planner context, reducing planning context overhead.

Task Memory. The task memory e_t maintains the query-specific context throughout agentic interactions. Specifically, it records the selected document skill and progressively accumulates information relevant to resolving the query, such as retrieved evidence, inspected pages or regions, intermediate results, and interaction history. As tool execution proceeds, these entries are continuously updated to reflect the evolving context of the current query.

As task memory contains detailed information relevant to solving the ongoing query, it provides the context required for subsequent tool planning. Therefore, the task memory e_t is fully exposed to the planner at each interaction step, allowing the agent to directly access accumulated evidence, inspected targets, and interaction history when determining the next action. Once the final output is produced, the task memory is discarded. This prevents query-specific context from carrying over to subsequent queries.

3.3 Document Skills

While the agentic interaction formulates a unified execution workflow for intelligent document processing, queries with different objectives require distinct interaction strategies and tool-use patterns. To accommodate these diverse objectives, DocClaw lets the agent select a document skill from a pre-defined document skill set based on its interpretation of the query objective. Each document skill provides task-specific guidance for subsequent planning and tool selection.

OCR-oriented Skill. The OCR-oriented skill is designed for accurate document transcription. Similar to pipeline-based methods (Zhang et al. 2026b; Wang et al. 2026), it first guides layout analysis to identify document regions and routes different region types to specific parsing tools, including text recognition, formula parsing, and table parsing. To further improve recognition quality, the skill encourages visual enhancement of low-quality regions and guides the agent to apply zoom, crop, and rotate operations before re-recognition. Together, these instructions allow the agent to adapt its processing strategy to the content and quality of regions through specialized parsing and visual enhancement.

DocQA-oriented Skills. DocQA-oriented skills target reliable question answering through multimodal search and evidence extraction. Since distinct document-based questions may require either localized evidence inspection or broader document-level coverage, we divide the question-answering policy into two complementary skills, the DocQA inspection skill and the DocQA enumeration skill. The inspection skill adopts a coarse-to-fine searching strategy that progressively narrows candidate pages, inspects relevant content, and extracts supporting evidence before answer generation. In contrast, the enumeration skill prioritizes broad document-level coverage for questions that require document-wide reasoning. The skill encourages the agent to maintain a sufficiently broad candidate page scope, thereby reducing the risk of insufficient page coverage in document-wide tasks such as occurrence counting and item grouping.

KIE-oriented Skill. The KIE-oriented skill targets reliable extraction of requested fields from document pages. The skill first guides direct field extraction from the page image. To mitigate errors from purely visual extraction, it then invokes OCR to obtain textual content and performs a second extraction from the recognized text. For fields with inconsistent predictions, the skill further specifies conflict resolution using the original page image as a reference to select the more reliable value. Overall, the skill combines visual extraction, OCR-based extraction, and image-grounded verification to improve the accuracy of structured field-value extraction.

4 Experiments

4.1 Experimental Setup

Benchmarks and Metrics. We evaluate DocClaw on three intelligent document processing tasks: optical character recognition (OCR), document question answering (DocQA), and key information extraction (KIE).

The OCR evaluation is conducted on OmniDocBench v1.6 (Ouyang et al. 2025), a document parsing benchmark containing 1,651 PDF pages. Following the official evaluation protocol, we report the overall score and category-specific metrics that cover different document modules. Specifically, Text Edit measures normalized edit distance for recognized text, Formula CDM evaluates formula recognition, and Table TEDS and Table TEDS-S evaluate table recognition with and without considering cell content, respectively. Reading Order Edit measures the correctness of

predicted reading sequences. The overall score aggregates performance across text, formula, and table modules.

For DocQA, we evaluate DocClaw on MMLongBench-Doc (Ma et al. 2024), a document question answering benchmark consisting of 135 PDF documents with an average of 47.5 pages per document. Adhering to the official evaluation protocol, raw responses are first normalized through LLM-based answer extraction using GPT-4.1 (OpenAI 2025) and evaluated by rule-based scoring against the ground-truth answers. Each question receives a score between 0 and 1 according to the match between the extracted prediction and the ground-truth answer, and the final accuracy is computed as the average score over all questions.

For KIE, we evaluate DocClaw on the KIE subsets of OCRBench (Liu et al. 2024) and OCRBench v2 (Fu et al. 2025). OCRBench contains 200 KIE samples formulated as single-field question answering, where predictions are evaluated by checking whether the ground-truth answer appears in the normalized output. OCRBench v2 contains 400 English and 400 Chinese KIE samples, both requiring structured key-value extraction. Following the official evaluation protocol, predictions are parsed into normalized field-value pairs and evaluated by field-level exact matching.

Implementation Details. We instantiate DocClaw with three multimodal foundation models as agent backbones: GPT-5.2, Gemini-2.5-Flash, and Gemini-2.5-Pro. For each backbone, the same model is used for the planner and model-based document processing tools unless otherwise specified.

The document processing toolset spans three categories: parsing, search, and analysis. Parsing tools recognize text, tables, formulas, and charts, primarily using PaddleOCR-VL-1.6 (Zhang et al. 2026b). Search tools support keyword search, text-based retrieval, image-based retrieval, and VLM-based visual search using BM25 (Robertson and Zaragoza 2009), ColBERT (Khattab and Zaharia 2020), ColPali (Faysse et al. 2025), and the backbone VLM, respectively. Analysis tools support layout analysis with PP-DocLayout v3 (Sun et al. 2025b), image understanding using the backbone VLM, and visual refinement operations such as zoom, crop, and rotate. Other model-based tools perform OCR quality inspection, evidence extraction, answer generation, and candidate verification using the same backbone as the planner. All experiments are conducted on a server equipped with an Intel Xeon Platinum 8369B CPU and two NVIDIA A100 80GB GPUs.

4.2 Main Results

Optical Character Recognition. We report OCR results on OmniDocBench v1.6 in Table 1(a). DocClaw consistently outperforms all baselines across all three backbone models. Particularly, DocClaw with Gemini-2.5-Pro obtains the best overall score of 96.45, compared with PaddleOCR-VL-1.6 at 96.34. Notably, DocClaw achieves the highest scores on Formula CDM, Table TEDS, and Table TEDS-S. These improvements are due to the targeted refinement of low-quality regions, where DocClaw applies visual enhancement operations such as zoom, crop, and rotate to improve readability and increase recognition accuracy in these regions.

(a) Optical Character Recognition								
Type	Method	Overall $\uparrow$	Text ^{Edit} $\downarrow$	Formula ^{CDM} $\uparrow$	Table ^{TEDS} $\uparrow$	Table ^{TEDS-S} $\uparrow$	Order ^{Edit} $\downarrow$	
General VLMs	GPT-5.2	86.59	0.114	88.21	82.95	87.93	0.193	
	Qwen3-VL-235B	89.78	0.063	92.55	83.07	86.75	0.166	
	Gemini-3-Pro	92.91	0.064	95.99	89.15	92.96	0.165	
Specialized VLMs	Youtu-Parsing	93.74	0.044	93.63	92.02	95.00	0.116	
	Qianfan-OCR	93.90	0.040	95.08	90.53	93.31	0.130	
	GLM-OCR	95.22	0.044	97.18	92.83	95.39	0.133	
	MinerU2.5-Pro	95.75	0.036	97.45	93.42	95.92	<u>0.120</u>	
	PaddleOCR-VL-1.6	96.34	0.033	97.53	<u>94.76</u>	<u>97.10</u>	0.128	
Document Agent	DocClaw w/ GPT-5.2	<u>96.40</u>	0.035	<u>98.01</u>	94.71	<u>97.10</u>	0.129	
	DocClaw w/ Gemini-2.5-Flash	96.38	<u>0.034</u>	<u>97.87</u>	<u>94.76</u>	97.13	0.129	
	DocClaw w/ Gemini-2.5-Pro	96.45	<u>0.034</u>	98.05	94.79	97.13	0.129	
(b) Document Question Answering								
Type	Method	ALL	TXT	LAY	CHA	TAB	FIG	UNA
General VLMs	GPT-4o	42.8	46.3	46.0	45.3	50.0	44.1	20.2
	Gemini-2.5-Flash	49.6	44.0	53.2	46.0	43.9	48.2	56.7
	Gemini-2.5-Pro	58.1	52.1	<u>62.1</u>	55.5	55.3	54.0	59.9
Specialized Pipelines	MDocAgent w/ GPT-4o	42.0	—	—	—	—	—	—
	DocAgent w/ GPT-4o	51.8	—	—	—	—	—	—
	DocDancer w/ Gemini-2.5-Pro	56.3	—	—	—	—	—	—
	DocDancer w/ GPT-5.2	57.0	—	—	—	—	—	—
	SimpleDoc w/ Gemini-2.5-Pro	56.6	48.4	54.8	55.7	56.1	52.5	59.7
	SimpleDoc w/ Claude-4-Sonnet	58.6	52.1	53.3	<u>58.3</u>	<u>62.4</u>	46.9	<u>66.5</u>
Document Agent	DocClaw w/ GPT-5.2	59.3	61.0	60.1	58.1	60.4	<u>55.8</u>	58.7
	DocClaw w/ Gemini-2.5-Flash	<u>61.0</u>	55.2	57.2	54.9	<u>63.0</u>	55.5	66.8
	DocClaw w/ Gemini-2.5-Pro	61.5	<u>56.4</u>	62.9	60.1	64.8	60.3	56.0
(c) Key Information Extraction								
Type	Method	ALL	OCRBench KIE	OCRBench v2 KIE (EN)	OCRBench v2 KIE (CN)			
General VLMs	GPT-5.2	69.8	88.0	76.1	45.4			
	Gemini-2.5-Flash	76.2	91.8	83.3	53.6			
	Gemini-2.5-Pro	77.0	<u>93.5</u>	83.3	54.3			
Specialized VLMs	Qianfan-OCR	84.0	88.0	81.6	82.3			
	GLM-OCR	39.7	60.0	29.3	29.1			
Document Agent	DocClaw w/ GPT-5.2	81.0	92.5	76.6	73.7			
	DocClaw w/ Gemini-2.5-Flash	<u>84.8</u>	<u>93.5</u>	<u>84.2</u>	76.8			
	DocClaw w/ Gemini-2.5-Pro	86.9	94.5	87.5	<u>78.7</u>			

Table 1: Main results across optical character recognition, document question answering, and key information extraction. Panels (a) to (c) report results on OmniDocBench v1.6, MMLongBench-Doc, and the KIE subsets of OCRBench and OCRBench v2, respectively. For document question answering, TXT, LAY, CHA, TAB, FIG, and UNA denote text, layout, chart, table, figure, and unanswerable questions. For key information extraction, ALL averages the results across OCRBench KIE and OCRBench v2 subsets. The best and second-best results for each metric are highlighted in bold and underlined, respectively.

Document Question Answering. As shown in Table 1(b), DocClaw consistently achieves superior performance across different backbone models on MMLongBench-Doc. With Gemini-2.5-Pro, DocClaw reaches the best overall score of 61.5, outperforming the Gemini-2.5-Pro baseline by 3.4 points and all specialized pipelines. These improvements stem from two key designs. First, compared with general VLMs, DocClaw leverages document processing tools to ac-

tively locate and inspect relevant content. Second, DocClaw iteratively explores the document and dynamically assesses whether the evidence accumulated in the document state is sufficient to answer the question, thereby efficiently guiding subsequent exploration based on the current document state.

Key Information Extraction. We report the KIE results in Table 1(c). DocClaw achieves competitive performance

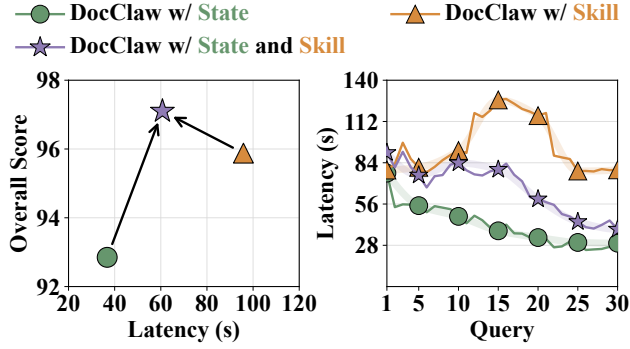


Figure 3: Ablation of document state and document skills. Left: overall score and average latency when using document state, document skills, or both. Right: latency across 30 consecutive queries on the same document.

across all backbone models, reaching the best overall score of 86.9 and outperforming the specialized Qianfan-OCR by 2.9 points. These improvements are attributed to the KIE document skill, which guides the agent to extract target field-value pairs through visual and OCR-based extraction, together with image-grounded verification to resolve inconsistent pairs.

4.3 Ablation Studies

Effect of Document State and Skills. We study the effects of document state and document skills on a manually constructed set of 30 randomly shuffled queries over the same document, using Gemini-2.5-Flash as the backbone. The set comprises 10 queries each for OCR, DocQA, and KIE. As shown in Figure 3, adding document skills to the state-enabled variant improves the overall score from 92.85 to 97.10, demonstrating the benefit of task-oriented guidance, at the cost of additional processing latency. In contrast, adding document state to the skill-enabled variant reduces the latency from 95.81s to 60.70s. The right panel shows that the latency of state-enabled variants decreases as more queries are processed, as accumulated document knowledge can be reused by subsequent queries to avoid repeated processing.

Effect of OCR Refinement. We analyze the contribution of refinement operations invoked during optical character recognition, including zoom, crop, and rotate. For pages involving multiple operations, we evenly distribute the performance improvement among all applied operations. As shown in Figure 4, refinement is triggered on 433 pages, with zoom applied most frequently to 354 pages, compared with 77 for crop and 41 for rotate. Zoom contributes most of the overall improvement, with an attributed gain of 0.23. It is particularly effective for formula recognition, where zooming improves the visibility of small mathematical symbols. Crop and rotate operations benefit table parsing by correcting region boundaries and orientations for more reliable recognition results.

Effect of KIE Verification. We also study the effect of OCR and verification operations in the KIE task on OCR-Bench v2, using Gemini-2.5-Pro as the vanilla VLM baseline. As shown in Figure 5, incorporating OCR improves the

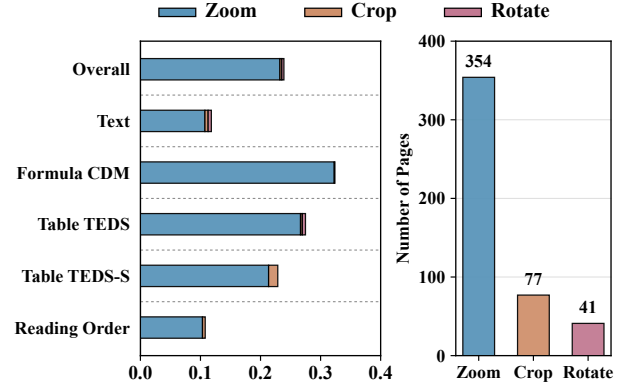


Figure 4: Ablation of OCR refinement. Left: performance improvement attributed to zoom, crop, and rotate operations. Right: number of pages refined by each operation.

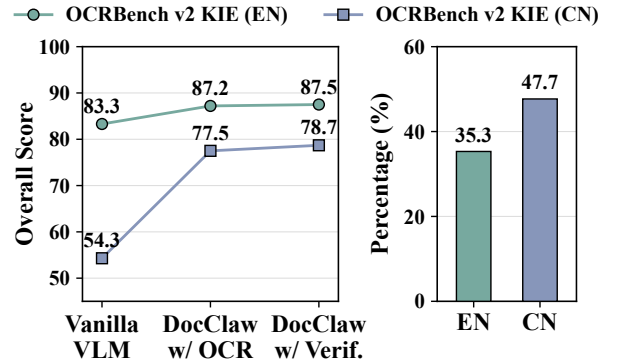


Figure 5: Ablation of OCR and verification for KIE. Left: KIE performance of the vanilla VLM and DocClaw with OCR and verification operations. Right: percentage of vanilla VLM extraction errors corrected by the verification operation.

KIE score from 83.3 to 87.2 on the English subset and from 54.3 to 77.5 on the Chinese subset, while verification further improves the scores to 87.5 and 78.7, respectively. Notably, verification corrects 35.3% and 47.7% of the extraction errors made by the vanilla VLM on the English and Chinese subsets. These results show that the verification operation can effectively correct erroneous field predictions.

5 Conclusion

We introduced DocClaw, a unified agentic system that formulates intelligent document processing as interactions between an agent and a document. DocClaw iteratively invokes processing tools under the guidance of document skills while maintaining a structured document state that organizes reusable document knowledge and task-specific interaction context. Extensive experiments demonstrate competitive performance against general-purpose VLMs and task-specific methods. Further ablation studies validate the effectiveness of document state and skills while demonstrating the contribution of task-specific refinement and verification operations.

References

- Appalaraju, S.; Jasani, B.; Kota, B. U.; Xie, Y.; and Manmatha, R. 2021. Docformer: End-to-end transformer for document understanding. In *Proceedings of the IEEE/CVF international conference on computer vision*, 993–1003.
- Chan, C.; Chen, W.; Su, Y.; Yu, J.; Xue, W.; Zhang, S.; Fu, J.; and Liu, Z. 2024. ChatEval: Towards Better LLM-based Evaluators through Multi-Agent Debate. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net.
- Dong, D.; Zheng, M.; Xu, D.; Luo, C.; Zhuang, B.; Li, Y.; He, R.; Wang, H.; Zhang, W.; Wang, W.; Wang, Y.; Xiong, X.; Zheng, A.; Zuo, X.; Ou, Z.; Gu, J.; Guo, Q.; Wu, J.; Yin, D.; and Shen, D. 2026. Qianfan-OCR: A Unified End-to-End Model for Document Intelligence. *CoRR*, abs/2603.13398.
- Duan, S.; Xue, Y.; Wang, W.; Su, Z.; Liu, H.; Yang, S.; Gan, G.; Wang, G.; Wang, Z.; Yan, S.; Jin, D.; Zhang, Y.; Wen, G.; Wang, Y.; Zhang, Y.; Zhang, X.; Hong, W.; Cen, Y.; Yin, D.; Chen, B.; Yu, W.; Gu, X.; and Tang, J. 2026. GLM-OCR Technical Report. *CoRR*, abs/2603.10910.
- Faysse, M.; Sibille, H.; Wu, T.; Omrani, B.; Viaud, G.; Hudelot, C.; and Colombo, P. 2025. ColPali: Efficient Document Retrieval with Vision Language Models. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net.
- Feng, H.; Wei, S.; Fei, X.; Shi, W.; Han, Y.; Liao, L.; Lu, J.; Wu, B.; Liu, Q.; Lin, C.; Tang, J.; Liu, H.; and Huang, C. 2025. Dolphin: Document Image Parsing via Heterogeneous Anchor Prompting. In Che, W.; Nabende, J.; Shutova, E.; and Pilehvar, M. T., eds., *Findings of the Association for Computational Linguistics, ACL 2025, Vienna, Austria, July 27 - August 1, 2025*, volume ACL 2025 of *Findings of ACL*, 21919–21936. Association for Computational Linguistics.
- Fu, L.; Kuang, Z.; Song, J.; Huang, M.; Yang, B.; Li, Y.; Zhu, L.; Luo, Q.; Wang, X.; Lu, H.; Li, Z.; Tang, G.; Shan, B.; Lin, C.; Liu, Q.; Wu, B.; Feng, H.; Liu, H.; Huang, C.; Tang, J.; Chen, W.; Jin, L.; Liu, Y.; and Bai, X. 2025. OCR-Bench v2: An Improved Benchmark for Evaluating Large Multimodal Models on Visual Text Localization and Reasoning. In Belgrave, D.; Zhang, C.; Montoya, L. N.; Lin, H.; Pascanu, R.; Koniusz, P.; Ghassemi, M.; Chen, N.; Ruíz, I. V. M.; and Loaiza-Bonilla, A., eds., *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2025, NeurIPS 2025, San Diego, CA, USA, December 2-7, 2025 / Mexico City, Mexico, November 30 - December 5, 2025*.
- Han, S.; Xia, P.; Zhang, R.; Sun, T.; Li, Y.; Zhu, H.; and Yao, H. 2025. MDocAgent: A Multi-Modal Multi-Agent Framework for Document Understanding. *CoRR*, abs/2503.13964.
- Ji, Y.; Xu, Z.; Liu, Z.; Chen, Z.; Zhang, Q.; Yang, Z.; Lin, J.; Gu, Y.; Yu, G.; and Sun, M. 2026. UNIKIE-BENCH: Benchmarking Large Multimodal Models for Key Information Extraction in Visual Documents. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 6331–6352.
- Khattab, O.; and Zaharia, M. 2020. ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT. In Huang, J. X.; Chang, Y.; Cheng, X.; Kamps, J.; Murdock, V.; Wen, J.; and Liu, Y., eds., *Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval, SIGIR 2020, Virtual Event, China, July 25-30, 2020*, 39–48. ACM.
- Lewis, P.; Perez, E.; Piktus, A.; Petroni, F.; Karpukhin, V.; Goyal, N.; Küttler, H.; Lewis, M.; Yih, W.; Rocktäschel, T.; Riedel, S.; and Kiela, D. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In Larochelle, H.; Ranzato, M.; Hadsell, R.; Balcan, M.; and Lin, H., eds., *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*.
- Li, G.; Hammoud, H.; Itani, H.; Khizbullin, D.; and Ghanem, B. 2023. CAMEL: Communicative Agents for "Mind" Exploration of Large Language Model Society. In Oh, A.; Naumann, T.; Globerson, A.; Saenko, K.; Hardt, M.; and Levine, S., eds., *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*.
- Li, M.; Lv, T.; Chen, J.; Cui, L.; Lu, Y.; Florencio, D.; Zhang, C.; Li, Z.; and Wei, F. 2021a. Trocr: Transformer-based optical character recognition with pre-trained models. *arXiv preprint arXiv:2109.10282*.
- Li, Y.; Qian, Y.; Yu, Y.; Qin, X.; Zhang, C.; Liu, Y.; Yao, K.; Han, J.; Liu, J.; and Ding, E. 2021b. Structext: Structured text understanding with multi-modal transformers. In *Proceedings of the 29th ACM international conference on multimedia*, 1912–1920.
- Liu, Y.; Li, Z.; Huang, M.; Yang, B.; Yu, W.; Li, C.; Yin, X.; Liu, C.; Jin, L.; and Bai, X. 2024. OCRBench: on the hidden mystery of OCR in large multimodal models. *Sci. China Inf. Sci.*, 67(12).
- Ma, Y.; Zang, Y.; Chen, L.; Chen, M.; Jiao, Y.; Li, X.; Lu, X.; Liu, Z.; Ma, Y.; Dong, X.; Zhang, P.; Pan, L.; Jiang, Y.; Wang, J.; Cao, Y.; and Sun, A. 2024. MMLONGBENCH-DOC: Benchmarking Long-context Document Understanding with Visualizations. In Globerson, A.; Mackey, L.; Belgrave, D.; Fan, A.; Paquet, U.; Tomczak, J. M.; and Zhang, C., eds., *Advances in Neural Information Processing Systems 37: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*.
- OpenAI. 2025. Introducing GPT-4.1 in the API. <https://openai.com/index/gpt-4-1/>.
- Ouyang, L.; Qu, Y.; Zhou, H.; Zhu, J.; Zhang, R.; Lin, Q.; Wang, B.; Zhao, Z.; Jiang, M.; Zhao, X.; Shi, J.; Wu, F.; Chu, P.; Liu, M.; Li, Z.; Xu, C.; Zhang, B.; Shi, B.; Tu, Z.; and He, C. 2025. OmniDocBench: Benchmarking Diverse PDF Document Parsing with Comprehensive Annotations. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025, Nashville, TN, USA, June 11-15, 2025*, 24838–24848. Computer Vision Foundation / IEEE.

- Robertson, S. E.; and Zaragoza, H. 2009. The Probabilistic Relevance Framework: BM25 and Beyond. *Found. Trends Inf. Retr.*, 3(4): 333–389.
- Schick, T.; Dwivedi-Yu, J.; Dessì, R.; Raileanu, R.; Lomeli, M.; Hambro, E.; Zettlemoyer, L.; Cancedda, N.; and Scialom, T. 2023. Toolformer: Language Models Can Teach Themselves to Use Tools. In Oh, A.; Naumann, T.; Globerson, A.; Saenko, K.; Hardt, M.; and Levine, S., eds., *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*.
- Sun, L.; He, L.; Jia, S.; He, Y.; and You, C. 2025a. DocAgent: An Agentic Framework for Multi-Modal Long-Context Document Understanding. In Christodoulopoulos, C.; Chakraborty, T.; Rose, C.; and Peng, V., eds., *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, EMNLP 2025, Suzhou, China, November 4-9, 2025*, 17701–17716. Association for Computational Linguistics.
- Sun, T.; Cui, C.; Du, Y.; and Liu, Y. 2025b. PP-DocLayout: A Unified Document Layout Detection Model to Accelerate Large-Scale Data Construction. *CoRR*, abs/2503.17213.
- Tang, Z.; Yang, Z.; Wang, G.; Fang, Y.; Liu, Y.; Zhu, C.; Zeng, M.; Zhang, C.; and Bansal, M. 2023. Unifying vision, text, and layout for universal document processing. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 19254–19264.
- Team, H. V.; Lyu, P.; Wan, X.; Li, G.; Peng, S.; Wang, W.; Wu, L.; Shen, H.; Zhou, Y.; Tang, C.; Yang, Q.; Peng, Q.; Luo, B.; Yang, H.; Zhang, X.; Zhang, J.; Peng, H.; Yang, H.; Xie, S.; Zhou, L.; Pei, G.; Wu, B.; Yan, R.; Wu, K.; Yang, J.; Wang, B.; Liu, K.; Zhu, J.; Jiang, J.; Linus, H.; and Zhang, C. 2025. HunyuanOCR Technical Report. *CoRR*, abs/2511.19575.
- Wang, B.; He, T.; Ouyang, L.; Wu, F.; Zhao, Z.; Chu, T.; Qu, Y.; Jin, Z.; Zeng, W.; Miao, Z.; Xu, B.; Niu, J.; Cai, M.; Qiu, J.; Zhang, Q.; Ma, D.; Sun, Y.; Dong, H.; Zhang, W.; Xiao, J.; Shi, J.; Liao, P.; Zhao, X.; Zhong, H.; Wei, L.; Yu, J.; Yang, J.; Li, W.; Wang, S.; Wu, Q.; Zhou, X.; Li, W.; Li, Z.; Tu, Z.; Wu, J.; Wu, L.; Xu, C.; Chen, K.; Zhang, W.; Qiao, Y.; Zhou, B.; Lin, D.; and He, C. 2026. MinerU2.5-Pro: Pushing the Limits of Data-Centric Document Parsing at Scale. *CoRR*, abs/2604.04771.
- Wu, Q.; Bansal, G.; Zhang, J.; Wu, Y.; Zhang, S.; Zhu, E.; Li, B.; Jiang, L.; Zhang, X.; and Wang, C. 2023. AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation Framework. *CoRR*, abs/2308.08155.
- Xu, Y.; Li, M.; Cui, L.; Huang, S.; Wei, F.; and Zhou, M. 2020. LayoutLM: Pre-training of Text and Layout for Document Image Understanding. In Gupta, R.; Liu, Y.; Tang, J.; and Prakash, B. A., eds., *KDD '20: The 26th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Virtual Event, CA, USA, August 23-27, 2020*, 1192–1200. ACM.
- Xu, Z.; Chen, Z.; Liu, Q.; Ji, J.; Hu, J.; Yu, Y.; Wan, J.; Tang, J.; and Li, Z. 2026. Enhancing Large Multimodal Models in Key Information Extraction via Scene-Aware Document Synthesis. *arXiv preprint arXiv:2607.04636*.
- Yao, S.; Zhao, J.; Yu, D.; Du, N.; Shafran, I.; Narasimhan, K. R.; and Cao, Y. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net.
- Ye, J.; Hu, A.; Xu, H.; Ye, Q.; Yan, M.; Xu, G.; Li, C.; Tian, J.; Qian, Q.; Zhang, J.; et al. 2023. Ureader: Universal ocr-free visually-situated language understanding with multimodal large language model. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2841–2858.
- Zhang, J.; Liu, Y.; Wu, Z.; Pang, G.; Ye, Z.; Zhong, Y.; Ma, J.; Wei, T.; Xu, H.; Chen, W.; Wang, Z.; Ji, Q.; Zhou, F.; Zhang, Q.; Hu, Y.; Liu, J.; Li, Z.; Zhang, Z.; Liu, Q.; and Bai, X. 2025. MonkeyOCR v1.5 Technical Report: Unlocking Robust Document Parsing for Complex Patterns. *CoRR*, abs/2511.10390.
- Zhang, Q.; Lv, X.; Wu, J.; Li, B.; Tao, Z.; Yan, G.; Zhang, H.; Wang, B.; Xu, J.; Mi, H.; and Zhang, W. 2026a. DocDancer: Towards Agentic Document-Grounded Information Seeking. *CoRR*, abs/2601.05163.
- Zhang, Z.; Liu, H.; Liang, S.; Zhang, Y.; Xiang, Y.; Liu, J.; Sun, T.; Lin, M.; Zhang, Y.; Zhou, C.; Gao, T.; Cui, C.; Liu, Y.; Yu, D.; and Ma, Y. 2026b. PaddleOCR-VL-1.6: Expanding the Frontier of Document Parsing with Under-Optimized Region Refinement and Progressive Post-Training. *arXiv:2606.03264*.